Yuzhao Chen

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Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in applied AI and human–AI collaboration. Experienced in building systems with large language models and vision–language models, applying controllable generation, multimodal reasoning, and agentic workflows to solve engineering problems.

Education

Ph.D. in Electrical and Computer Engineering — Aug. 2023 – Aug. 2027 (Expected)
Purdue University — West Lafayette, IN

M.S. in Electrical and Computer Engineering — Sep. 2021 – Jun. 2023
University of California San Diego — La Jolla, CA

B.S. in Mathematics — 2019–2021
University of Liverpool — Liverpool, UK

B.S. in Applied Mathematics — 2017–2019
Xi'an Jiaotong-Liverpool University — Suzhou, China

Selected Projects

JustShape — Gesture-Guided 3D Parametric Modeling [C.4] **2025**

- Introduced co-speech gestures as a spatial input modality for AI-assisted parametric modeling, conveying dimensions, directions, and shapes that are difficult to specify through language alone.
- Developed an agentic workflow that fuses speech and hand-tracking data into explicit modeling parameters, with a Unity-based augmented reality interface for expressing design intent and viewing generated 3D models.
- Reduced geometric error by 40.1% versus spoken descriptions alone and interaction time by 47.3% versus drawing sketches while describing the design.

Canvas3D — Spatially Guided Image Generation [C.2] **2025**

- Built an interactive 3D canvas that turns language descriptions into editable scenes, enabling users to directly arrange objects and control spatial relationships in multi-object image generation.
- Used large language models to interpret scene descriptions and organize 3D objects, with controls for object placement, viewpoint, human poses, and lighting.
- Encoded scene configurations as depth maps, pose data, and lighting parameters to guide image composition; evaluated support for spatial expression and generation through controlled tasks and open-ended exploration.

AmIWrite — AI Tutoring with Speech and Handwriting [C.3] **2025**

- Built a multimodal AI tutor that interprets handwritten equations and solution steps, combining spoken explanations with written annotations for personalized, ongoing mathematical problem-solving support.
- Grounded feedback in specific equations and steps, adding circles, arrows, correction marks, and worked steps directly to handwritten solutions in sync with spoken explanations.
- In a 40-participant study, reduced NASA-TLX mental demand scores from 43.1 to 29.5 and increased System Usability Scale scores from 63.4 to 78.4 compared with a text-only tutoring interface.

SketchConcept — Sketch-Guided Product Concept Design [C.5] **2024**

- Built an AI design tool combining sketch-based visual forms with functional requirements expressed through language, supporting product concept generation, decomposition, and local design iteration.
- Combined semantic segmentation with vision–language reasoning to map parts in generated concept images to their intended design functions, connecting visual representations with functional requirements for part-level editing.
- Enabled users to select parts, revise functional descriptions, and regenerate corresponding regions; evaluated function-guided design exploration and recomposition through application cases and a two-stage user study.

Technical Skills

Programming: Python, MATLAB

Tools & Platforms: Linux, Docker, PTC Creo

ML Frameworks & Libraries: PyTorch, Transformers, Diffusers, OpenCV, TensorFlow, LLaMA-Factory

Honors and Fellowships

Led a project that received the **Honorable Mention Award** at ACM CHI 2026.

Purdue Mechanical Engineering **Perry Teaching Fellowship**, 2026

Publications, Preprints, and Submitted Manuscripts

C: Conference Papers **J**: Journal Articles **PP**: Preprints **S**: Manuscripts Under Review
Tags: [Technical Methods & Capabilities] [Interaction & Human–AI Research] [Research Contexts & Applications]

[C.5] SketchConcept: Sketching-based Concept Composition for Product Design using Multimodal Large Language Model
Runlin Duan*, Chenfei Zhu*, ***Yuzhao Chen****, Dizhi Ma, Jingyu Shi, Yichen Hu, Ziyi Liu, Karthik Ramani
ACM DIS 2026 [Controllable Generation] [Multimodal AI] [AI for Engineering]

[C.4] JustShape: Exploring Co-Speech Gestures for Multimodal LLM-Powered 3D Parametric Modeling
Runlin Duan*, ***Yuzhao Chen****, Yichen Hu, Ziyi Liu, Chenfei Zhu, Xiyun Hu, Dizhi Ma, Xinyi Wang, Karthik Ramani
ACM CHI 2026 (Honorable Mention Award) [Multimodal Interaction] [Mixed Reality] [AI for Engineering]

[C.3] AmIWrite: Exploring Scalable One-on-One Handwriting-Based Tutoring for Mathematical Problem-Solving with an LLM-Powered AI Tutor
Ziyi Liu*, ***Yuzhao Chen****, Haoyu Ji, Runlin Duan, Zhengzhe Zhu, Xiyun Hu, Kylie Peppler, Karthik Ramani
ACM CHI 2026 [Multimodal Interaction] [Visual Grounding] [Human–AI Collaboration]

[C.2] Canvas3D: Empowering Precise Spatial Control for Image Generation with Constraints from a 3D Virtual Canvas
Yuzhao Chen*, Runlin Duan*, Rahul Jain, Yichen Hu, Chenfei Zhu, Jingyu Shi, Karthik Ramani
ACM IUI 2026 [Controllable Generation] [Spatial Computing] [Human–AI Collaboration]

[C.1] DesignFromX: Empowering Consumer-Driven Design Space Exploration through Feature Composition of Referenced Products
Runlin Duan*, Chenfei Zhu*, ***Yuzhao Chen***, Yichen Hu, Jingyu Shi, Karthik Ramani
ACM DIS 2025 [Controllable Generation] [Human–AI Collaboration] [AI for Engineering]

[J.1] ConceptVis: Exploring and Visualizing Design Concepts With Large Language Models Using Interactive Knowledge Graph
Runlin Duan, Nachiketh Karthik, ***Yuzhao Chen***, Jingyu Shi, Rahul Jain, Maria Yang, Karthik Ramani
ASME JCISE 2026 [Human–AI Collaboration] [Interactive Visualization] [AI for Engineering]

[PP.2] Investigating Creativity in Humans and Generative AI Through Circles Exercises
Runlin Duan, Shao-Kang Hsia, ***Yuzhao Chen***, Yichen Hu, Ming Yin, Karthik Ramani
arXiv preprint, 2025 [Generative AI] [Human–AI Comparison]

[PP.1] Taxonomic Class Incremental Learning
Yuzhao Chen*, Zonghuan Li*, Zhiyuan Hu, Nuno Vasconcelos
arXiv preprint, 2023 [Continual Learning] [Hierarchical Learning] [Computer Vision]

[S.5] SituatAR: Cross-Aspect Authoring of Embodied Human-Agent Interactions in Augmented Reality
Chenfei Zhu*, ***Yuzhao Chen****, Bingyi Liu, Yichen Hu, Ziyi Liu, Xinyi Wang, Dizhi Ma, Shao-Kang Hsia, Karthik Ramani
Submitted to ACM CHI 2027 [Agentic AI] [Mixed Reality] [Multimodal Interaction]

[S.4] InterReason: Sharing and Shaping Situated Reasoning with AR Agents
Chenfei Zhu, Bingyi Liu, Xiyun Hu, ***Yuzhao Chen***, Xinyi Wang, Karthik Ramani
Submitted to ACM CHI 2027 [Human–AI Collaboration] [Agentic AI] [Mixed Reality]

[S.3] SOUND: Sensemaking Over Unexpected Noise for Diagnosis of Everyday Machines with Multimodal LLMs
Shao-Kang Hsia, Shengmin Ding, Nathan Maharaj Kharwanlang, Runlin Duan, Ziyi Liu, Chenfei Zhu, Dizhi Ma, ***Yuzhao Chen***, Xinyi Wang, Xiyun Hu, Karthik Ramani
Submitted to ACM CHI 2027 [Multimodal AI] [AI for Engineering] [Human–AI Collaboration]

[S.2] RealiTTy: Situated Multi-Ball Training for Table Tennis in Mixed Reality
Dizhi Ma, Chenfei Zhu, Shao-Kang Hsia, ***Yuzhao Chen***, Xiyun Hu, Nishant Vasan, Karthik Ramani
Submitted to IEEE VR 2027 [Mixed Reality] [Motion Tracking] [Computer Vision]

[S.1] ComposeVR: An In-Situ Authoring System for Compositional Object Behaviors in Virtual Reality
Shao-Kang Hsia, Chenfei Zhu, Runlin Duan, Xinyi Wang, Xiyun Hu, Dizhi Ma, ***Yuzhao Chen***, Karthik Ramani
Submitted to IEEE VR 2027 [Agentic AI] [Spatial Computing] [Human–AI Collaboration]

** denotes equal contribution.*